

Importance Sampling Using Multilabel Radial Basis Classification for Composite Power System Reliability Evaluation

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Abstract—Resource additions to electric power grids are planned in advance to maintain reliability of electric power supply. This is achieved by performing reliability studies for calculating reliability indices at the planning stage to ensure that the required levels of reliability will be met. This article proposes a new state classification approach to calculate power system reliability indices using a combination of multilabel radial basis function (MLRBF) networks and importance sampling (IS) within the framework of Monte Carlo simulation process. Multilabel classification algorithms are different from single-label approaches, in which each instance can be assigned to multiple classes. This characteristic gives MLRBF the capability to classify composite power system states (success or failure), at the bus as well as system level. Bus level indices provide useful reliability information for locational reliability, which allows more rational and equitable distribution of resources. MLRBF classification does not require optimal power flow (OPF) analysis; however, OPF is required for the training and cross-entropy optimization phases. This article shows that the scope and computational efficiency to evaluate reliability indices can be significantly increased if the proposed MLRBF classifier is used together with well-known variance reduction technique of IS. As the classifier is trained to recognize the reliability status of a system state, it can also be used in operational planning when a number of scenarios need to be evaluated in a short time for their ability to satisfy load. The proposed method is illustrated using the IEEE reliability test system for different load levels. The outcomes of case studies show that MLRBF algorithm together with IS provides excellent classification accuracy in reliability evaluation while substantially reducing computation time and enhancing the scope of evaluation.

Index Terms—Composite power system reliability evaluation, cross entropy (CE) algorithm, importance sampling (IS), Monte Carlo simulation (MCS), multilabel radial basis function (RBF) learning algorithm.

NOMENCLATURE

A. Abbreviations

ANN	Artificial neural networks.
AIR	Artificial immune recognition.
CE	Cross entropy.
CMCS	Crude or basic Monte Carlo simulation.
COV	Coefficient of variation.
IS	Importance sampling.
MLL	Multilabel learning.
MLKNN	Multilabel K -nearest neighbor.
MLRBF	Multilabel radial basis function.
LHS	Latin hypercube sampling.
LOL	Loss of load.
LOLP	Loss of load probability.
KNN	K -nearest neighbor.
OPF	Optimal power flow.
RBF	Radial basis function.
RTS	Reliability test system.
MRTS	Modified reliability test system.
SVM	Support vector machine.

B. Indices and Parameters

m	Index of samples.
n	Index of generation buses.
q	Index of system buses.
β	Parameter for coefficient of variance.

C. Variables

G_{mn}	Generation for sample m at bus n .
P_{mq}	Estimation of ANN for sample m at bus q .
T_{mq}	Target variable for sample m at bus q .
$Y_m(q)$	Conditional probability for sample m at bus q .

I. INTRODUCTION

RELIABLE electric power grids are essential for societal wellbeing, industrial productivity, and national security. Therefore, resource planning is done by power utilities to ensure that required level of reliability will be met when resources are installed. This is typically done by independent system operators (ISOs) or utilities by performing reliability planning studies. These studies use reliability evaluation techniques for estimating reliability indices at the planning stage and ensure that these

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indices satisfy the specified criterion. This process of simulation and prediction is important at the planning stage as by its very nature; this infrastructure once built is not possible to make changes in an economical fashion.

Power system reliability evaluation techniques can be classified into various categories but simulation-based methods are capable of incorporating more details of the complex systems and, thus, make the models more reflective of reality [1]. Therefore, these are generally methods of choice for reliability analysis of composite power systems, including generation and transmission resources. The Monte Carlo simulation (MCS) based techniques, including, importance sampling (IS) and Latin hypercube sampling, currently appear to be the preferred methods to estimate the reliability indices for composite power systems [2].

Although MCS-based techniques are powerful in their ability to incorporate details, their drawback is that they can take extensive computational time especially while evaluating the high reliability systems, which have very low probability of failure to estimate and, thus, failures are rare events. The high computational time originates from two sources. The first is the extensive number of samples required to converge to a reasonable accuracy measured by coefficient of variation (COV) and the second is time required to evaluate sampled states since an optimal power flow (OPF) needs to be performed to characterize each state and OPF is a time-consuming procedure.

For reducing the number of samples, several approaches have been investigated but cross entropy (CE) based IS has been shown as a good solution [3]. One of the early applications of this idea in power systems can be found in [4] and [5]. Recently, the CE method has been successfully implemented for reliability evaluation of composite power systems [6], [7]. The results obtained from these studies show that CE-optimized IS provides a significant speed-up in MCS. However, despite more efficient sampling, the other drawback of needing OPF for characterizing each sampled state is still a bottle neck.

It should be mentioned at this point that research has been previously done for increasing computational efficiency by using artificial intelligence and machine learning. These researches include population-based intelligence search methods as an alternative to search for meaningful states to increase the computational efficiency of these simulation methods [8]–[15]. Pattern classification techniques have been explored to reduce the computational burden associated with OPF to classify system states as success or failure. Artificial neural network based classifiers [16], artificial immune recognition system [17], or least squares support vector machine based classifiers [18] are some of the successfully implemented examples of these techniques. These methods are, however, directed toward identifying the system state (failure–success) by utilizing the chosen classification method and increase the computational efficiency of reliability analysis by reducing the OPF burden. These approaches manage to classify successful system states with high accuracy and reduce the computational burden required for reliability analysis. However, their application has been for system level classification rather than at the bus level. The bus-level reliability indices, however, are very important as they provide locational

reliability information. The bus-level reliability indices help in identifying locations of poor reliability and, thus, promote distribution of resources to provide benefits of reliability in a more equitable manner. This article provides a method to use machine learning techniques for the bus-level evaluation, in addition to system level, thus overcoming the weakness of the previous methods. This more comprehensive approach will promote the acceptance of machine learning methods for power system reliability evaluation. The proposed approach uses algorithms based on multilabel learning.

Multilabel classification is a type of learning where each sample is associated with multiple labels, providing a capability for calculation of bus-level indices. The multilabel algorithms can be categorized into two main groups, problem transformation and algorithm adaptation methods [19]. The objective of transformation methods is to handle a multilabel classification problem by converting it into a single-label classification problem. Binary reverse method [19] or pairwise method [20] are examples for this approach. The algorithm adaptation methods on the other hand are designed from some specific single-label learning algorithms to solve multilabel classification problems directly. MLKNN [21], neural networks based MLL [22], or decision trees [23] are some examples of this group.

This article, thus, proposes a new approach by combining the MLRBF classifier [24] and IS approach for reliability evaluation of composite power systems both at the system as well as bus level. In the proposed method, the required samples are obtained from a distorted probability distribution (IS) and characterization of those states is done by the MLRBF classifier without requiring OPF except during the training phase. The efficiency of the proposed approach is demonstrated on the 24 bus IEEE reliability test system (RTS) and modified RTS (MRTS). It should also be mentioned that RBF has been used in some other applications in power systems, such as probabilistic power flow [25]–[28]. The objective of the approach proposed in this article, however, is quite different and is to classify states without running OPF.

The remainder of this article is organized as follows. In Section II, the mathematical background of the MLRBF is briefly provided for composite system reliability analysis. Section III gives brief information about IS and the CE algorithm for power systems. The application of the proposed method is explained in steps in Section IV. Case studies on the RTS are discussed in Section V. Finally, conclusions are given in Section VI.

II. MULTILABEL CLASSIFICATION FOR POWER SYSTEM RELIABILITY EVALUATION

RBF is one of the most popular among neural network classification methods. RBF neural networks are generally comprised of two layers of neurons. In RBF, each hidden neuron (basis function) in the first layer is associated with a prototype vector, whereas each output neuron corresponds to a possible class. Usually training an RBF neural network is handled using a two-stage procedure. In the first layer, the basic functions are learned by performing clustering analysis on training instances,

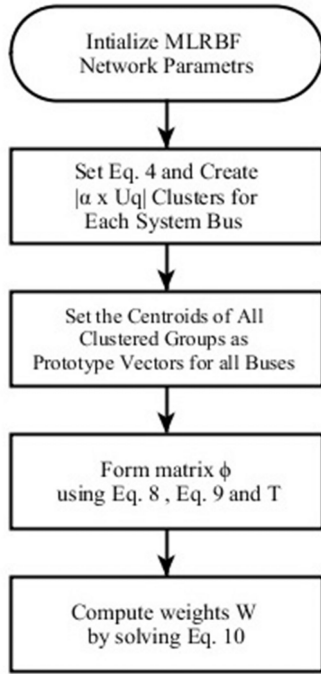


Fig. 1. Flowchart describing training phase of MLRBF.

whereas weights are optimized by solving a linear problem in the second layer. Comprehensive descriptions of RBF neural networks are available in [29].

In this section, first, a general formulation of the composite power system reliability evaluation parameters for MLRBF classification is explained, later, the application of the proposed method is described. Finally, a flowchart of MLRBF is provided for a clearer understanding in Fig. 1.

A. General Definitions for Multilabel Learning in Power System Reliability Evaluation

In this article, the total generation capacities and the total demand for each bus of the composite system are taken as input parameters for the MLRBF classifier. So, generation and demand information for each bus in the system is considered as an element of the input matrix I for every sample (instance) m as described

$$I_{\text{input}} = \begin{bmatrix} G_{11} - D_{11} & G_{12} - D_{12} & G_{1N} - D_{1N} \\ G_{21} - D_{21} & G_{22} - D_{22} & G_{2N} - D_{2N} \\ G_{M1} - D_{M1} & G_{M2} - D_{M2} & G_{MN} - D_{MN} \end{bmatrix} \quad (1)$$

where N is the number of the buses and M is the total number samples in the input matrix.

The status of the state information for each bus of the system for M different samples is stored in a target matrix T for the purpose of training the MLRBF classifier, which is described as

$$T = \begin{bmatrix} S_{11} & S_{12} & S_{1Q} \\ S_{21} & S_{22} & S_{2Q} \\ S_{M1} & S_{M2} & S_{MQ} \end{bmatrix} \quad (2)$$

where Q is the number of the load buses in the system and S is the status information of bus q . For defining status of buses, “−1” is used for “success states” and “1” for “failure states” for the corresponding bus.

P_{out} contains the failure probability for each bus of the composite system for each sample M as the output for this classifier, which is described as

$$P_{\text{out}} = \begin{bmatrix} P_{11} & P_{12} & P_{1Q} \\ P_{21} & P_{22} & P_{2Q} \\ P_{M1} & P_{M2} & P_{MQ} \end{bmatrix}. \quad (3)$$

Next, the steps in the training and testing procedure are explained in the following section.

B. Explanation of the MLRBF Classification Procedure

It is necessary to describe some related parameters before starting the explanation where

- m index of current sample of the total M samples;
- i_m input vector for sample m ;
- q bus index of total Q buses of the system;
- Y_m state of bus q in sample m so

$$Y_m(q) = \begin{cases} 1 & \text{(failure) where } T_{iq} = 1 \\ 0 & \text{(success) where } T_{iq} = -1. \end{cases}$$

Let $I = R^d$ be the input space and $Q = \{1, 2, \dots, Q\}$ be the finite set of Q possible classes. Given a multilabel training dataset $D\text{Set} = \{(i_m, Y_m) | 1 \leq m \leq M\}$, where $i_m \in I$ is a single instance and $Y_m \subseteq Q$ is label set associated with i_m .

In this article, k means clustering is applied for each class $q \in Q$ on the set of instances U_q with label q , which is described as

$$U_q = \{i_m | (i_m, Y_m) \in D\text{Set}, q \in Y_m\}. \quad (4)$$

In the next step, k_q number of clustered groups are formed for class q and the j th centroid ($1 \leq j \leq k_q$) is regarded as a prototype vector c_j^q of basis function $\alpha_j^1(\cdot)$. It should be noted that k_q is taken as a fraction of the total number of instances in U_q , which is described as α .

As each output neuron of the MLRBF, a neural network is related to a possible class, weights between the hidden, and the output layer can be shown as

$$W = [w_{jq}]_{(K+1) \times Q}. \quad (5)$$

Here, $K = \sum_{q=1}^Q k_q$ is the total number of prototype vectors retained in the hidden layer. The weight matrix W can be learned by minimizing the sum-of-squares error function described as

$$E = \frac{1}{2} \sum_{m=1}^M \sum_{q=1}^Q (Y_q(i_m) - T_q^m)^2 \quad (6)$$

where T_q^m represents the output of i_m on the q th class, which takes the values of +1 if $q \in Y_i$ and −1 otherwise. So, the output

of i_m for the q th class can be calculated as

$$y_q(i_m) = \sum_{j=0}^Q w_{jq} \phi_j(i_m). \quad (7)$$

In this article, the basis function a_j is represented with the widely used Gaussian style activation

$$\phi_j(i_m) = \exp\left(-\frac{\text{dist}(i_m, c_j)^2}{2\sigma_j^2}\right). \quad (8)$$

Here, $\text{dist}(i_m$ and $c_j)$ calculates the distance between i_m and the j th prototype vector c_j with the usual Euclidean distance algorithm.

The smoothing parameter σ is shown as

$$\sigma = \left(\frac{\sum_{p=1}^{K-1} \sum_{r=p+1}^K \text{dist}(c_p, c_r)}{\frac{K(K-1)}{2}} \right). \quad (9)$$

Differentiating the error function (6) with respect to w_{jq} and setting the derivative to zero will result in

$$(\phi^T \phi) W = \phi^T T. \quad (10)$$

In (10), $\phi = [\phi_{mj}]_{m \times [K+1]}$ with elements $\phi_{mj} = \phi_j(i_m)$, $W = [w_{jq}]_{Q \times [K+1]}$, and $T = [t_{mq}]_{m \times Q}$ with elements $t_{mq} = t_q^m$.

III. IS AND CE ALGORITHM

IS is one of the most successful variance reduction techniques used in reliability evaluation of composite power systems [30]. IS changes the probability density function of occurrences by emphasizing certain values of a random variable, which have greater impact, when compared with other values, on the estimation process of a target quantity.

Consequently, values, which have more importance, are sampled more often and the variance of the estimator is reduced faster. The goal of IS is to select a probability density function different from the original to minimize the variance of samples [6]. To be able to obtain the maximum performance from IS, the selected probability density function should be equal or close to optimum $f_{\text{optimum}}(\cdot)$, which is initially unknown. The CE method can be utilized for estimating the optimal, or at least near optimal, reference parameters by minimizing the distance between the original sampling density and the optimal sampling density $f_{\text{optimum}}(\cdot)$ iteratively.

Detailed technical information can be obtained from [31]. In the following sections, general definitions regarding IS are presented and later the implementation of the CE method for power system reliability evaluation is described.

A. General Description of IS

Consider a power system with G_N generating stations. Also, assume that the system has J identical and independent units, each with a capacity G_{nj} for each of N stations. Let u_{nj} be a vector that contains original probability of unavailability of all generation units in the system. Under this assumption, the analytical problem of evaluating the loss of load probability

(LOLP) index can be described by the following equation:

$$\text{LOLP} = \frac{1}{M} \sum_{i=1}^M H(X_i) \quad (11)$$

where X_i represents i th sample of M total samples and H represents the test function that takes value of 1 if sample X_i has loss of load and 0 otherwise. Under these assumptions, IS can be applied to the system by using a new unavailability vector v_{nj} to calculate H_{IS}

$$\text{LOLP}_{IS} = \frac{1}{M_{IS}} \sum_{i=1}^{M_{IS}} H_{IS}(X_i) W(X_i; m_{IS}; u; v). \quad (12)$$

The expression $W(X_i; m_{IS}; u; v)$ is called likelihood ratio that represents a necessary correction in the sampling process because of changed unavailability vector v . In this article, $W(X_i; m_{IS}; u; v)$ is calculated by

$$W(X_i; m_{IS}; u; v) = \frac{\prod_{j=1}^{N_G} (1 - u_j)^{x_j^G} (u_j)^{1-x_j^G}}{\prod_{j=1}^{N_G} (1 - v_j)^{x_j^G} (v_j)^{1-x_j^G}} \quad (13)$$

where x_j^G represents the availability of generation unit j . The main problem in this process is determining optimal v values to minimize computation time. In this article, CE algorithm is utilized for this purpose, which is explained in the following section.

B. CE Algorithm

In this section, CE algorithm to determine optimal unavailability (v) values for each generation unit is described. Detailed information about CE can be found in [3]–[5].

The algorithm used in this article converges to optimal v parameters using an iterative procedure. During each iteration, v parameters are updated by using predefined number of system state samples. The CE algorithm contains following six main steps. While optimal v parameters are estimated in steps 1–4, loss of load indices are calculated with the IS-MCS in steps 5 and 6.

- 1) Define the initialization parameters as sample size used for each iteration N (e.g., 50 000) and multilevel parameter p (e.g., between 0.01 and 0.1). Define $v_0 = u$, $t = 1$, and $\phi = L_{LMAX}$ where v represents updated unavailability vector, t iteration number, ϕ stopping criteria for performance function, and L_{LMAX} maximum peak load of the system.
- 2) Generate a set of random samples of states $X_1, X_2, \dots, X_N$ from the densities $f(\cdot, v_{t-1})$. Evaluate the performance of selected states $S = [S_1, S_2, \dots, S(X_N)]$ according to the selected performance function. Sort the performances of the states in an increasing order so that $S_{[1]} \leq S_{[2]} \leq \dots \leq S_{[XN]}$ then compute the performance of state (p) quantile of the performances $S_{[(p)XN]}$.
- 3) Set the $\phi_t = S_{[(1-p)XN]}$ provided that ϕ_t is less than ϕ otherwise set $\phi_t = \phi$. Evaluate the indicator function $H(X_i)$ such that $H(X_i) = 1$ if $S(X_i) > \phi_t$ otherwise $H(X_i) = 0$.

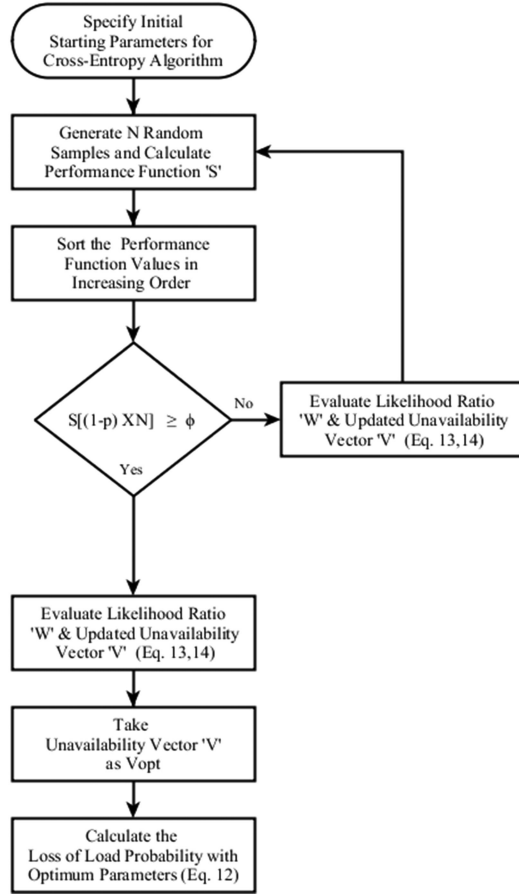


Fig. 2. Flowchart of IS process via CE algorithm.

- 4) Use the sample from step 2 to update the new unavailability vector

$$v_{tj} = 1 - \frac{\sum_{i=1}^{X_N} H_t(X_i) W_{i,t-1} X_{ij}}{\sum_{i=1}^{X_N} H_t(X_i) W_{i,t-1}} \quad (14)$$

where

$$W_{i,t-1} = W(X_i; u; v_{t-1}). \quad (15)$$

- 5) If $\phi_t = \phi$, the criteria have been satisfied and the optimal parameters has been found, otherwise increment the iteration count as $t = t + 1$ and go back to the step 2.
 6) Calculate the loss of load indices using (12) by using the optimal parameters defined in step 5.

A flowchart is provided in Fig. 2 for a clear understating of the CE method.

IV. APPLICATION PROCEDURE

This article proposes a new method for calculating reliability indices of composite power systems by combining multilabel classifiers and the IS technique within the framework of MCS. Generally, the most time-consuming part of composite system reliability evaluation is the OPF analysis. This article proposes use of a faster multilabel classifier instead of the OPF analysis after proper training. The proposed method basically replaces

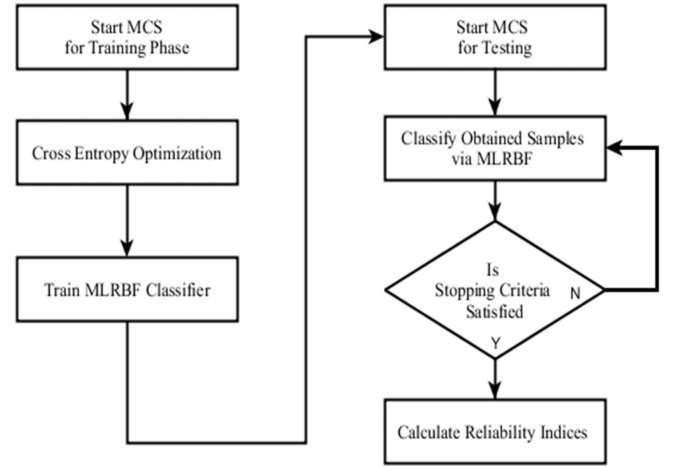


Fig. 3. Overall diagram of the proposed method.

OPF by the classifier and, thus, can be applied while using either nonsequential or sequential MCS. However, the nonsequential approach is chosen to illustrate performance in this article because of its simpler architecture. A benchmark created with the crude MCS (CMCS) analysis is also provided for comparison purpose.

The first step in applying the combination of IS and ML classification is determining the optimum unavailability vector via the CE method, as described in Section III. After the optimum unavailability vector is created, the multilabel classifier is trained and tested to use state-space classification for evaluating the reliability indices of the composite power system. Detailed implementation of the multilabel classifier is described in the sections below. Complete flowchart of the proposed methodology is presented in Fig. 3.

A. Training Process

Training data samples for this article are created through MCS. The generation and demand information for each bus of the selected sample states is used to create the input vector while the status of each bus for those states is recorded as the target vector, as shown in steps 1 and 2. To increase the training performance, the input vector variables are normalized between -1 and 1 . In this step, the failure and nonfailure status observed for each bus are labeled as 1 and -1 , respectively. It should also be noted that an equal amount of success and failure states is chosen to create the training dataset to prevent overtraining.

B. Testing or Index Evaluation Process

In the reliability indices calculation phase, MCS with IS is used for generating random samples and those states are classified by the multilabel classifier until the simulation reaches a previously determined stopping criterion. The stopping criteria for this article is defined as the COV to represent the estimated

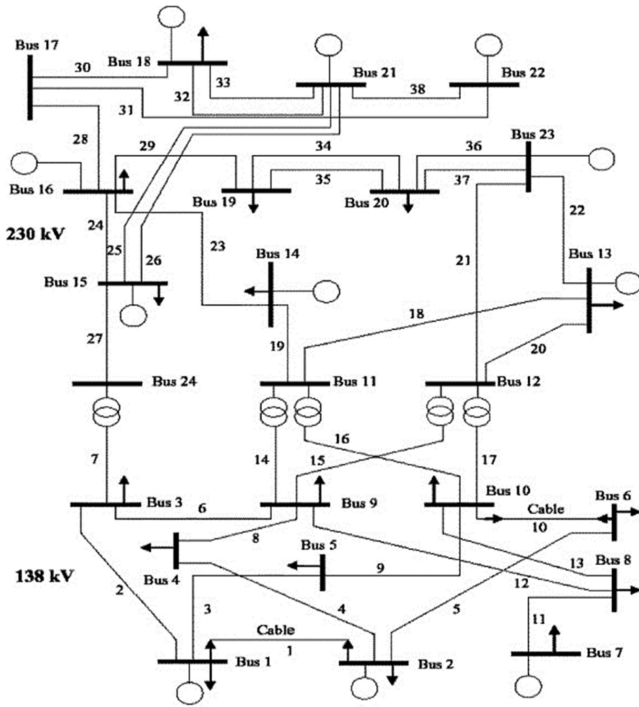


Fig. 4. Overall schematic of IEEE RTS 79 [33].

uncertainty. The COV is described by

$$\beta = \frac{\sqrt{V(E(F))}}{E(F)}. \quad (16)$$

In this equation, $V(E(F))$ represents the variance of the estimated value $E(F)$.

Finally, reliability indices are calculated and a comparison is made with the results obtained from crude MCS and IS benchmarks. Reliability indices are evaluated based on LOLP. Definition of LOLP is given as

$$\text{LOLP} = \frac{\text{Total Failure}}{\text{Total Number of Samples}}. \quad (17)$$

V. CASE STUDIES AND SIMULATION RESULTS

The IEEE RTS and MRTS are chosen for demonstration of the proposed method. The RTS consists of 24 buses, 38 transmission lines, and 32 generation units (distributed among ten buses). The RTS has a total installed capacity of 3405 MW and a load of 2850 MW at its annual peak [32]. The RTS is depicted in Fig. 4 [33]. MRTS has also similar topology but in this scenario generation capacity of each bus doubled and all load in the system increased in ratio of 1.8.

Three case studies are conducted to demonstrate the performance of the proposed method. In the first two case studies, the load level is considered constant at its annual peak for RTS and MRTS, respectively. In the last case study, the hourly load data of RTS are divided into five different load levels by considering their occurrence probabilities similar to [34]. After the application procedure described in the previous sections is

TABLE I
COMPARISON OF OVERALL PERFORMANCE ANALYSIS
AND CPU TIME SPENT IEEE RTS

Algorithm	Success States	Failure States	LOLP $\times 10^{-2}$	CPU Time (Sec $\times 10^3$)
CMCS	100459	9284	8.46	5.49
CEIS	18786	1748	8.51	1.03
ML-CEIS	21305	1988	8.53	0.069

completed, the performance comparison of the proposed multilabel classifier with the CMCS is conducted after the calculation of the reliability indices for all system buses. It should be noted that the straightforward MCS (CMCS) is used as the benchmark for comparing accuracy as well as efficiency of the proposed method. This is because the standard or crude Monte Carlo is an accepted method for power system reliability calculations and is used in commercial software, such as GE MARS, which is used by many organizations, such as New York ISO and New England ISO. Since system losses caused by transmission line failures are much lower than the ones occurring from generation unit failures, the states of the transmission lines are considered as available at all times. The capacities of transmission lines are, however, considered.

The initial parameters of the CE optimization are selected as $N = 50\,000$ (sample size) and $p = 0.05$ (multilevel parameter).

All simulations of this article are performed using the MATLAB (2012) platform on a PC with Intel Core i7-4510 CPU (~ 2.6 GHz), 16 GB memory. It should be noted that the results presented below are the average of ten simulations. It should also be noted that simulation accuracies and given times are average of many different simulations.

A. Constant Load Level for RTS

In this case study, the load level of the RTS is considered to be constant at its peak value of 2850 MW. To train the MLRBF classifier, 6000 samples are selected, 3000 of them are success states and the rest are failure states. The clustering rate α is chosen as 0.25 for this case study, which means the number of clusters created is equal to one fourth of the total failures for each bus, as described in (6).

After the training of the MLRBF classifier is completed, MCS is simulated until COV reaches 1% for all simulation types as specified for the stopping criteria in the testing phase.

In this process, a total of 109 743 samples were obtained with 100 459 successes and 9284 failures characterized. The comparison of the results is shown in Tables I and II.

The simulation results for the overall system classification performance and time comparison of the MLRBF classifier on the RTS are presented in Table I. As explained earlier, the results from CMCS are considered the bench mark. The simulation results for the bus-level classification performance are stated in Table II. In this article, the performance comparison is made based on the LOLP and overall simulation time.

Table II shows that the proposed method also shows a reasonably accurate classification for characterizing the failed bus states of the RTS.

TABLE II
COMPARISON OF CLASSIFICATION PERFORMANCE AT BUS-LEVEL
BASED ON LOLP FOR IEEE RTS

Location	CMCS (LOLP) $\times 10^{-2}$	CEIS (LOLP) $\times 10^{-2}$	ML-CEIS (LOLP) $\times 10^{-2}$
Bus 1	2.67	2.71	2.82
Bus 2	2.69	2.74	2.84
Bus 3	0.00	0.00	0.00
Bus 4	0.00	0.00	0.00
Bus 5	0.87	0.84	0.89
Bus 6	0.03	0.05	0.06
Bus 7	4.61	4.72	4.75
Bus 8	0.59	0.6	0.68
Bus 9	0.00	0.00	0.00
Bus 10	0.02	0.01	0.04
Bus 11	N/A	N/A	N/A
Bus 12	N/A	N/A	N/A
Bus 13	0.43	0.47	0.52
Bus 14	0.20	0.24	0.29
Bus 15	0.00	0.00	0.00
Bus 16	0.05	0.1	0.15
Bus 17	N/A	N/A	N/A
Bus 18	0.94	1.1	0.81
Bus 19	0.05	0.12	0.02
Bus 20	3.53	3.59	3.76
Bus 21	N/A	N/A	N/A
Bus 22	N/A	N/A	N/A
Bus 23	N/A	N/A	N/A
Bus 24	N/A	N/A	N/A

Table I shows that the MLRBF classification method can compute the overall LOLP of the RTS with a small fraction of error and the computational time required to evaluate the reliability indices can be significantly reduced by the proposed MLRBF – IS combination when compared to the standard MCS methods.

B. Constant Load Level for MRTS

In this case study, the load level of the MRTS is considered constant at its peak value of 5130 MW. To train the MLRBF classifier 10 000 samples are selected, 5000 of them are success states, and the rest are failure states. The clustering rate α is decided as 0.25.

After the training of the MLRBF classifier is completed, MCS is simulated until COV reaches 1% for all simulation types as specified for the stopping criteria in the testing phase.

In this process, a total of 198400 samples were obtained with 188871 successes and 9529 failures characterized. The comparison of the results is shown in Tables III and IV.

The simulation results for the overall system classification performance and time comparison of the MLRBF classifier on the RTS are presented in Table III. The simulation results for the bus-level classification performance are stated in Table IV. Performances are compared based on LOLP and total simulation time.

TABLE III
COMPARISON OF OVERALL PERFORMANCE ANALYSIS
AND CPU TIME SPENT MRTS

Algorithm	Success States	Failure States	LOLP $\times 10^{-2}$	CPU Time (Sec $\times 10^3$)
CMCS	188871	9529	4.84	10.36
CEIS	35211	1793	4.85	2.41
ML-CEIS	38912	1992	4.86	0.17

TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCE AT MRTS
BUS-LEVEL BASED ON LOLP

Location	CMCS (LOLP) $\times 10^{-2}$	CEIS (LOLP) $\times 10^{-2}$	ML-CEIS (LOLP) $\times 10^{-2}$
Bus 1	0.47	0.50	0.58
Bus 2	0.17	0.21	0.27
Bus 3	0.00	0.02	0.00
Bus 4	0.10	0.04	0.07
Bus 5	0.21	0.19	0.25
Bus 6	0.96	1.05	1.17
Bus 7	0.61	0.64	0.71
Bus 8	0.21	0.21	0.21
Bus 9	0.01	0.01	0.01
Bus 10	0.12	0.11	0.16
Bus 11	N/A	N/A	N/A
Bus 12	N/A	N/A	N/A
Bus 13	0.78	0.82	0.89
Bus 14	2.45	2.51	2.63
Bus 15	0.00	0.00	0.00
Bus 16	0.38	0.34	0.41
Bus 17	N/A	N/A	N/A
Bus 18	0.35	0.45	0.27
Bus 19	0.15	0.19	0.22
Bus 20	0.19	0.23	0.28
Bus 21	N/A	N/A	N/A
Bus 22	N/A	N/A	N/A
Bus 23	N/A	N/A	N/A
Bus 24	N/A	N/A	N/A

Table III shows that the MLRBF classification method can compute the overall LOLP of the MRTS with an acceptable accuracy; however, it is clear that the computational time required to evaluate the reliability indices dramatically reduced by the proposed MLRBF – IS combination when compared to the standard MCS methods.

Table IV shows that the proposed method also shows a reasonably accurate classification for characterizing the failed bus states of the MRTS.

C. Hourly Varying Load Levels

In this case study, the load level of the system is chosen randomly from the original load data of the RTS. There are 8736 different load levels specified in the annual hourly load values of the RTS. As in the first case study, samples used for training the MLRBF classifier are obtained through MCS.

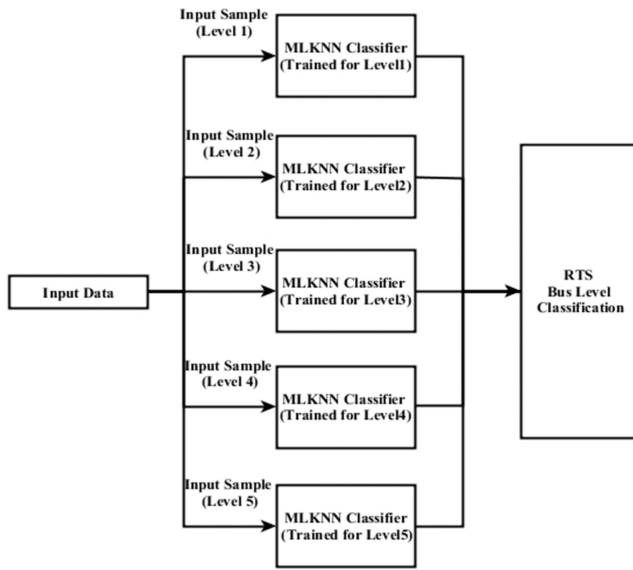


Fig. 5. Overall diagram of the proposed classifier for variable load levels.

TABLE V
COMPARISON OF OVERALL PERFORMANCE ANALYSIS AND
CPU TIME SPENT FOR VARYING LOAD LEVELS

Algorithm	Success States	Failure States	LOLP $\times 10^{-3}$	CPU Time (Sec $\times 10^3$)
CMCS	7433754	9125	1.20	376.690
CEIS	473232	1740	1.23	23.710
ML-CEIS	496277	1817	1.23	1.110

Since the size of the classifier is one of the most critical factors affecting the time for classification, multiple classifiers are trained for different load levels to tackle this problem efficiently instead of training one large network. For this purpose, five different thresholds are defined based on the available power. For each load level, a unique classifier is trained. In the testing stage, a decision tree is used to determine which classifier should be used for the classification of every random sample. The description of the algorithm used in this case study is given in Fig. 5.

For this case study, each load level is trained by 10 000 samples obtained through MCS sampled with the optimal unviability vector obtained through the CE process. Similar to the first case study, the clustering rate α is selected as 0.25 for each classifier.

After the training is completed, MCS is run until the COV reaches 1% as in the first case study. In this process, a total of 7 442 879 samples were obtained by CMCS with 7 433 754 successes and 9 125 failures characterized. The comparison of the classification performance and the simulation time for the results is presented in Tables V and VI.

The comparison of the overall classification performance and the simulation times for the proposed method is given in Table V. It is clear from Table V that the proposed method can classify failure states of the RTS in the multiload level case with a performance close to the benchmark methods. It is also observed

TABLE VI
COMPARISON OF CLASSIFICATION PERFORMANCE AT BUS-LEVEL
BASED ON LOLP FOR VARYING LOAD LEVELS

Location	CMCS (LOLP) $\times 10^{-5}$	CEIS (LOLP) $\times 10^{-5}$	ML-CEIS (LOLP) $\times 10^{-5}$
Bus 1	28.06	28.7	34.4
Bus 2	15.35	16.43	19.51
Bus 3	0	0.0	0
Bus 4	0	0	0
Bus 5	8.04	8.42	9.12
Bus 6	0.49	0.43	0.19
Bus 7	54.95	54.02	56.72
Bus 8	2.31	1.90	3.42
Bus 9	0	0	0
Bus 10	0	0	0
Bus 11	N/A	N/A	N/A
Bus 12	N/A	N/A	N/A
Bus 13	3.58	3.72	5.97
Bus 14	3.28	3.10	4.51
Bus 15	0	0	0
Bus 16	0.58	0.99	1.2
Bus 17	N/A	N/A	N/A
Bus 18	10.81	12.26	21.51
Bus 19	2.18	2.21	3.68
Bus 20	60.47	61.28	69.85
Bus 21	N/A	N/A	N/A
Bus 22	N/A	N/A	N/A
Bus 23	N/A	N/A	N/A
Bus 24	N/A	N/A	N/A

in Table V that the proposed method provides a significant saving in terms of computation time.

Table VI shows the performance of proposed method for bus-level classification. It can be observed from the results that the proposed method can compute the bus LOLP with a reasonable error.

VI. CONCLUSION

Reliability studies are an integral part of planning of electric power grids [35]. Most commonly used method for this purpose is the MCS that is flexible but can become computationally inefficient and, thus, become a hindrance when a large number of studies are to be performed to achieve optimal planning. To overcome this deficiency, a new method has been presented in this article to evaluate reliability indices for composite power systems. The proposed method uses an MLRBF classifier to identify the status of the system as well as buses without using an OPF analysis during MCS. The effectiveness of the proposed method is demonstrated using the IEEE RTS and MRTS.

As can be observed from the results, the MLRBF classifier can classify the loss of load states with excellent accuracy. Rate of classification discrepancy of some bus states may be a bit more with low frequency of failure at those locations. The main reason for this is the lack of adequate samples in the training dataset. The performance of the proposed method can be increased by adding more samples to the training dataset as a natural outcome.

The contributions of this article can be summarized as follows.

- 1) There are several advantages of the proposed approach compared to the previous methods. The most important is the ability to reduce the time for bus and system reliability analysis substantially as is shown in the three different case studies. This is achieved by focusing on the elements that are the main sources of computational burden. First, the sample size is reduced by using IS. Then, time for characterization of system state in each sample is reduced by using the classifier instead of OPF.
- 2) An important contribution of the proposed method, compared to the previous ones, is its ability to classify at the bus level in addition to the system level. The bus level indices are very important as they provide window on locational reliability, which ensures more equitable and fare distribution of resources and pricing.
- 3) An advantage of reducing the classification time is that more samples can be characterized without much additional effort and, thus, a finer convergence can be obtained leading to higher accuracy and consistency of results.
- 4) The computational efficiency plays an important role in performing sensitivity studies where results may need to be obtained repeatedly by varying certain parameters and, thus, computational time becomes critical if the method is to be accepted by the users. The information on locational reliability obtained through the bus-level indices can help in fair pricing of reliability of power supply at different locations in the system. The ability to compute bus-level indices in addition to system-level indices thus provides a better acceptance of these methods in power system reliability evaluation community.

From a broader perspective, although the classifier is used in this article for composite power system reliability evaluation, it can be used anywhere where state classification needs to be done to indicate whether there is loss of load or not. For example, this could become useful in operational planning where many scenarios need to be examined for their ability to satisfy load. It thus opens a whole new set of possibilities for further research and development.

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